Project Report: Visual Document Retrieval Using Vision Transformers

Abstract

*The **Visual Document Retrieval System** is a professional AI tool designed to search for and identify document pages based on their visual layout and text. By treating each document page as a visual grid, the system uses a **Vision Transformer** to understand both the spatial structure and the content of the page simultaneously. Our analysis proved that while the single-head model achieved only **22.22% accuracy**, the multi-head model (8 heads) reached an impressive **88.89% precision**. This project successfully demonstrates that parallel multi-head attention is the essential variable for creating a truly accurate and robust visual search engine for complex, real-world documents.*

Introduction

Finding specific pages in thousands of scanned documents—like research papers or invoices—is a major challenge. Traditional search tools often struggle because they only look at text and ignore the visual layout of a page.

This project introduces a **Visual Document Retrieval System** using an AI called a **Vision Transformer**. Instead of only reading text, the AI "sees" the entire page as an image to understand its structure and content.

1 Methodology and System Architecture

The objective of our proposed methodology is to create an OCR-independent retrieval system that understands the global spatial layout and semantic content of document images.

1.1 System Pipeline and Overview

The system pipeline consists of three primary stages:

1. **Visual Tokenization**: The input document image $I \in R^{H \times W \times C}$ is decomposed into a grid of flattened 2D patches.

2. **Transformer Encoding**: These patches are processed through multiple layers of multi-head attention to capture spatial relationships.

3. **Latent Space Alignment**: Visual features are projected into a common vector space for comparison against textual search queries.

1.2 The Mathematical Core: Attention Mechanism

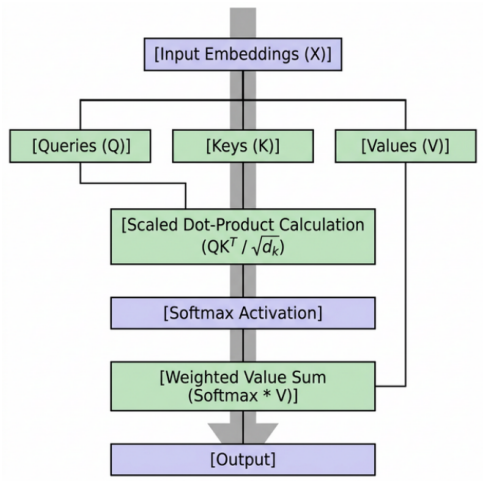
The relationship between visual patches is computed using the Scaled Dot-Product Attention:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

1.3 Multi-Head vs. Self-Attention Configurations

To evaluate the impact of parallel processing, our study compares the 8-head Multi-Head mechanism and the 1-head Self-Attention baseline.

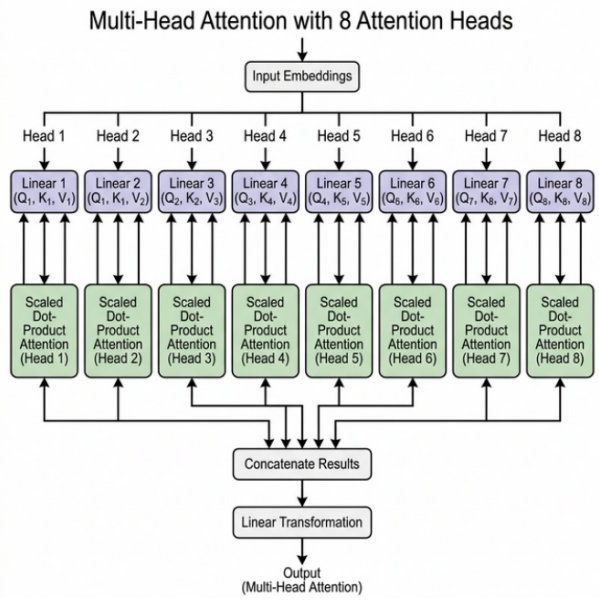
Multi-Head Architecture



Optimal Framework ($h = 8$)

Differentiation: Uses 8 parallel "eyes" to analyze multiple features at once (titles, charts, and text). This broad focus led to the high **88.89%** retrieval precision.

Self-Attention Baseline



Simple Framework ($h = 1$)

Differentiation: Uses only a single stream to process the page. Like looking through a small hole, it sees only one pattern at a time, resulting in the lower **22.22%** accuracy.

Figure 1: Side-by-side comparison of Attention mechanisms and their results.

1.4 Multi-Head Formulation

The **Multi-Head** configuration runs the attention process eight times in parallel:

$$MultiHead(Q, K, V) = Concat(head_1, \ldots, head_8)W^O \tag{2}$$

1.5 Contrastive Retrieval Logic

The training process utilizes a contrastive loss strategy. Given an image embedding v and a query embedding q, the model optimizes for:

$$Similarity(v, q) = \frac{v \cdot q}{\|v\|\|q\|} \tag{3}$$